

The figure below compares key characteristics across all four Swedish price zones over the 2015-2025 sample period. The top two panels show monthly mean price levels in raw and log scale. The third panel plots price volatility measured as the monthly mean of 24-hour rolling standard deviations. The bottom panel shows the forecasted wind production share of total production on a yearly basis. Wind penetration has increased across all zones, but SE1 and SE2 exhibit markedly higher and faster-growing shares.

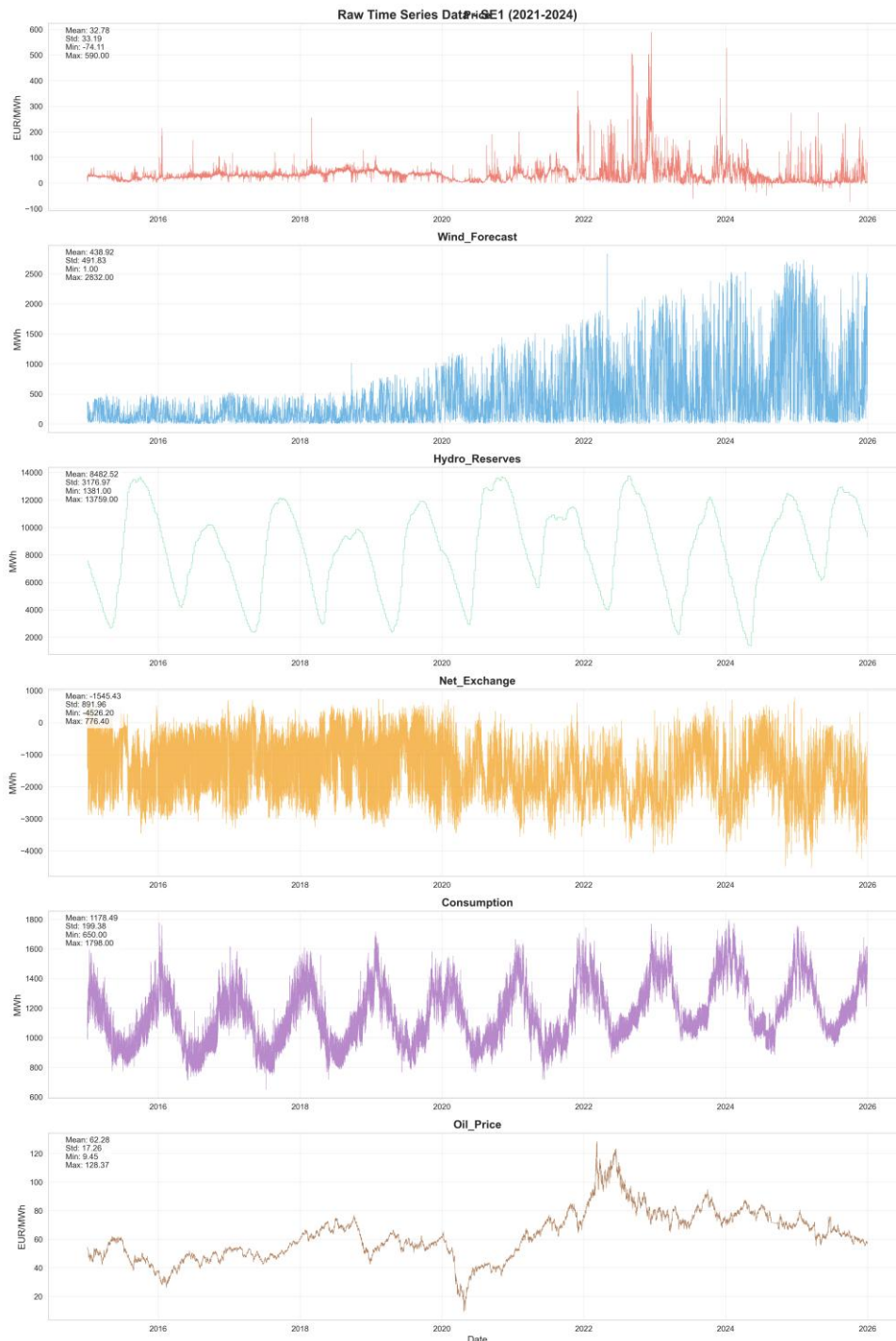
Zone Comparison — SE1-SE4 (2015-01-01 to 2025-12-31)



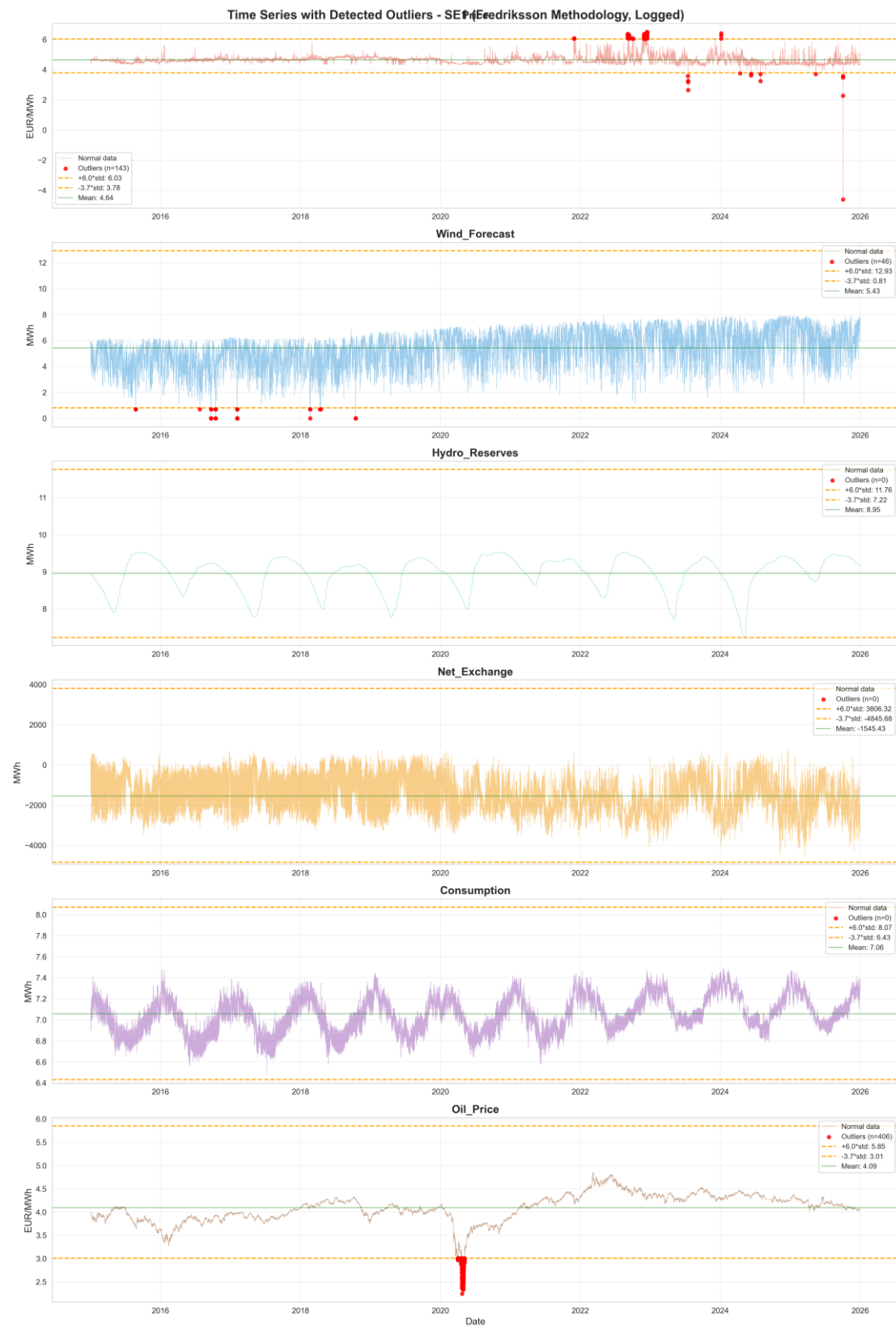
# SE1

The figure below presents the raw time series for all variables included in the analysis, shown here for SE1. The plots display hourly observations spanning

2015–2025 for electricity spot price (EUR/MWh), day-ahead wind power forecast (MWh), hydro reservoir levels (MWh), net exchange (MWh), consumption (MWh), and crude oil price (EUR/MWh). Each panel reports the series mean, standard deviation, minimum, and maximum. These plots are shown for a single zone as the variable structure is identical across SE1–SE4; the key differences between zones emerge in the regression analysis.

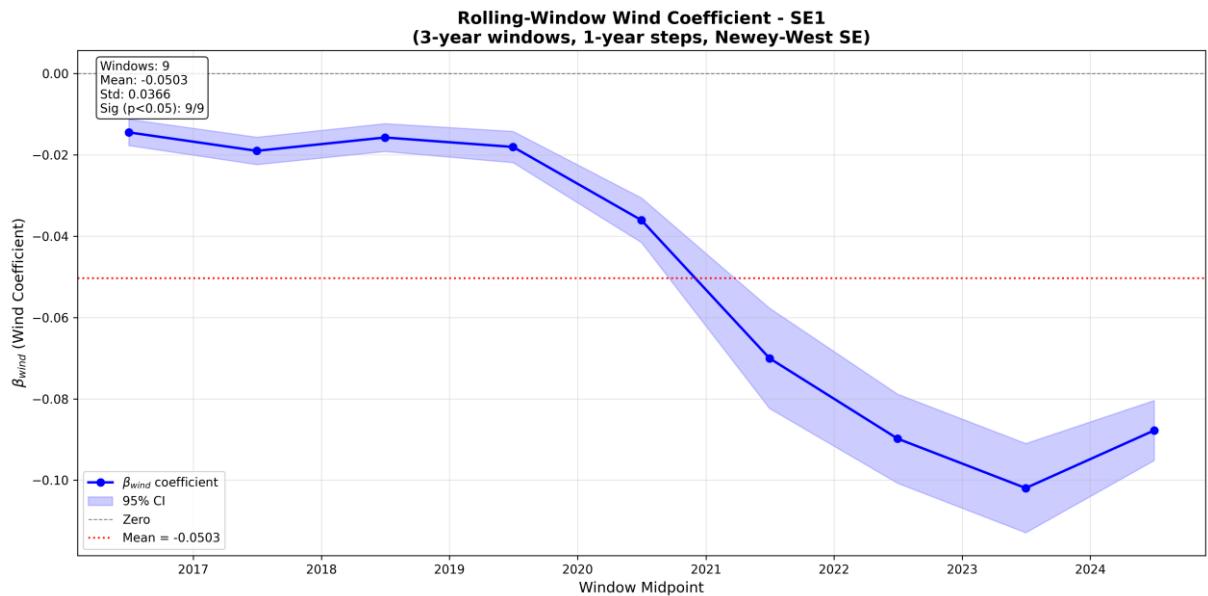


Below are the same series but logged and with outliers marked

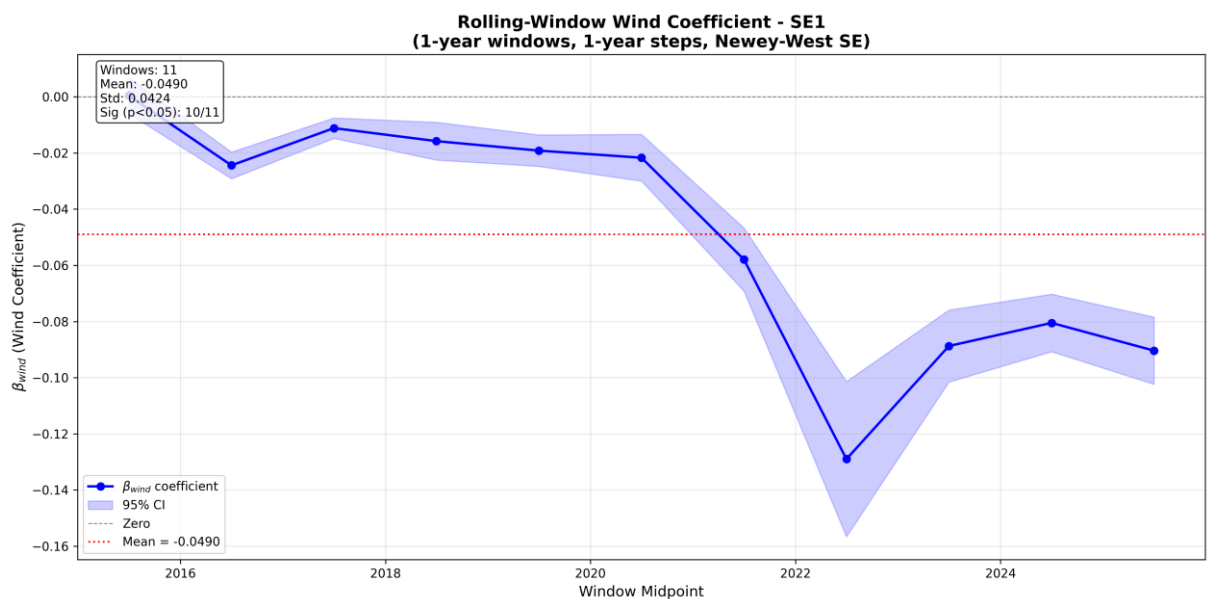


To examine how the wind power coefficient evolves over time, we re-estimate the regression model over rolling sub-samples. Three configurations are used: a 3-year window with 1-year steps, a 1-year window with 1-year steps, and a 1-year window with 1-month steps. Wider windows yield smoother estimates; narrower, frequently stepped windows capture finer dynamics at the cost of greater noise.

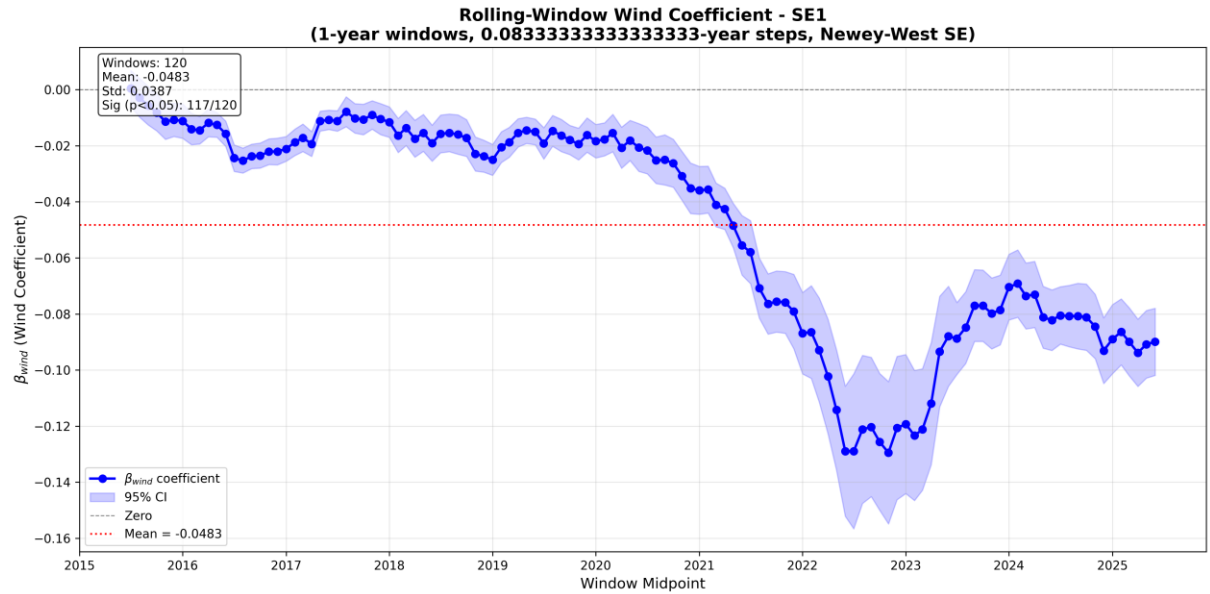
- 3 year, overlapping, shift 1 year:



- 1 year, no overlapping, shift 1 year:



- 1 year, overlapping, shift by 1 month



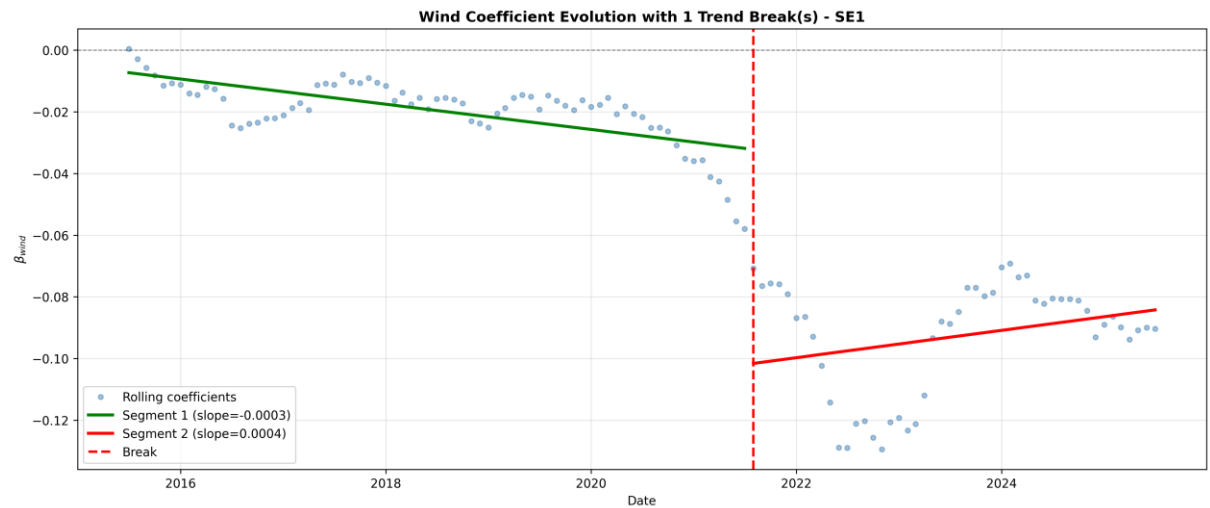
## Structural Break Analysis

To formally test whether the relationship between wind power and electricity prices undergoes significant shifts over time, we apply a sequential trend break analysis to the rolling coefficient series. First, the rolling wind coefficients are treated as a time series indexed by their window midpoints. For each candidate number of breaks  $m = 0, 1, 2, \dots$ , we fit a segmented linear regression where each segment has its own intercept and slope. The optimal placement of  $m$  break points is determined by minimizing the total residual sum of squares (RSS) across all segments, subject to a minimum segment length constraint (10% of the total number of windows). This is solved via a recursive search over all feasible break locations, trimming 15% from each endpoint to avoid spurious breaks near the boundaries. Model selection proceeds in two stages. First, the Bayesian Information Criterion (BIC) is computed for each specification, penalizing model complexity to guard against overfitting. The model with the lowest BIC is selected as the preferred specification. Second, sequential F-tests compare each successive pair of models ( $m$  vs  $m+1$  breaks), testing whether the additional break significantly reduces the RSS. A break is considered statistically justified only if the F-test rejects the null of no additional break at the 5% level.

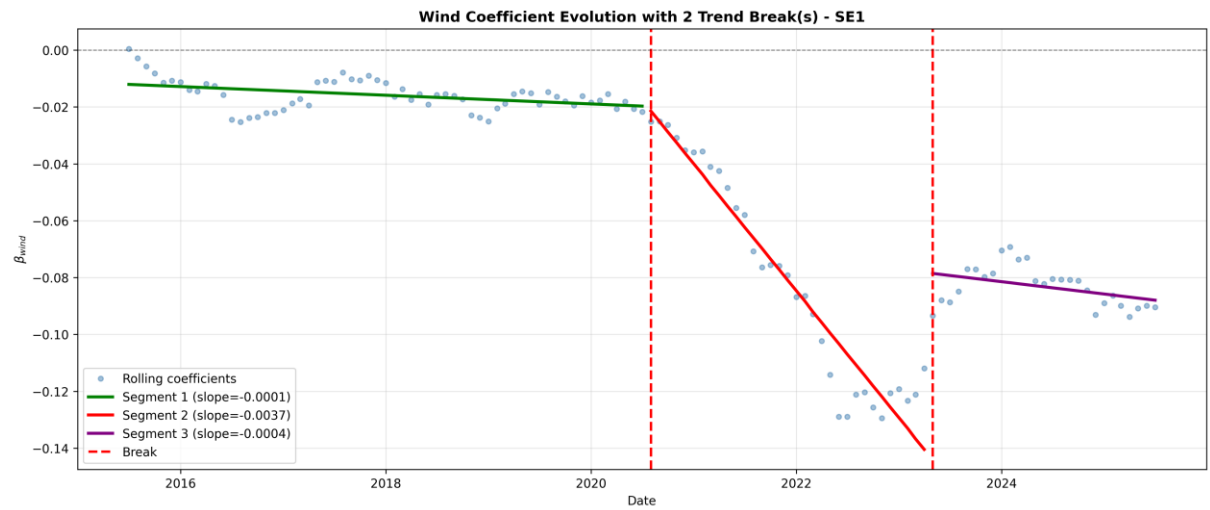
The figures present results for 1, 2, and 3 allowed breaks, showing the fitted trend segments and their slopes alongside the rolling coefficient estimates.

The break dates indicated by the dashed vertical lines correspond to the points where the trend in the wind coefficient changes most sharply.

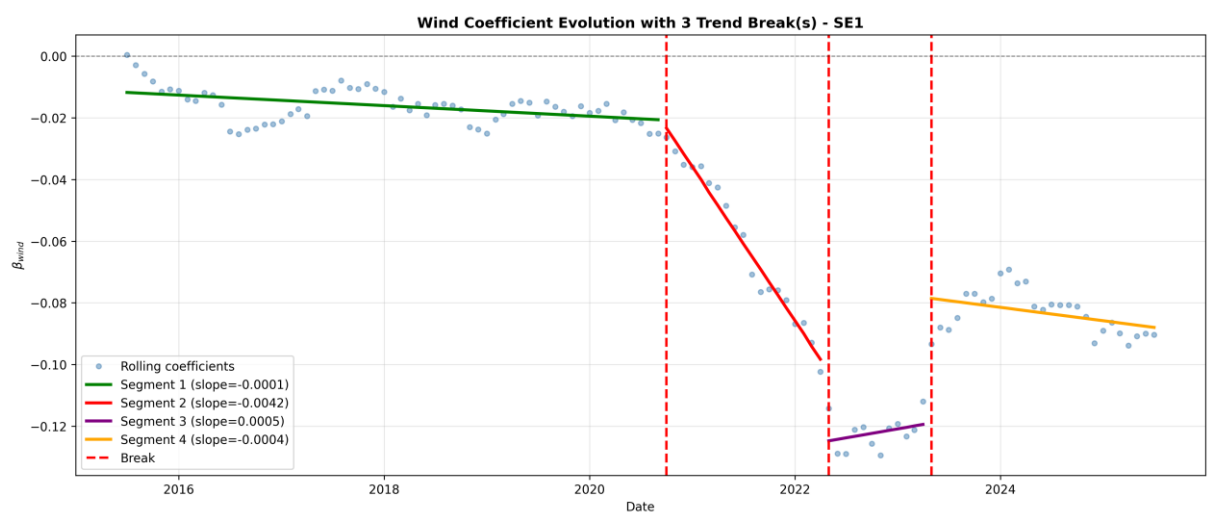
- 1 break allowed



- 2 breaks allowed

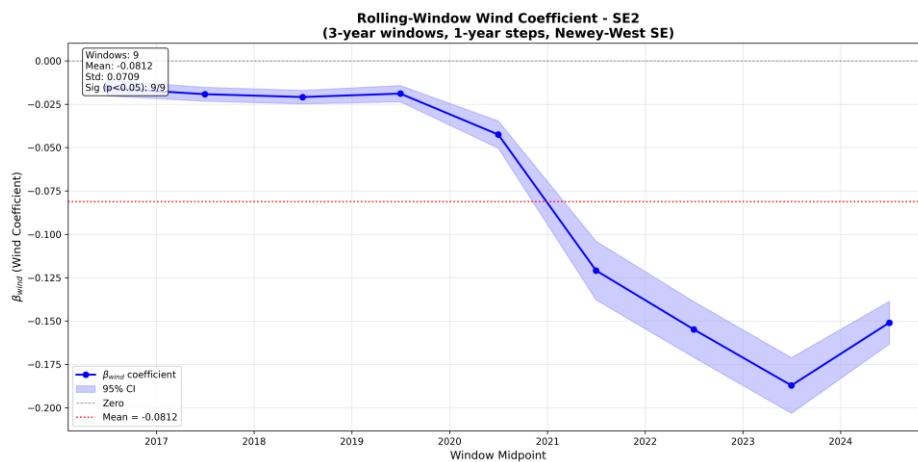


- 3 breaks allowed

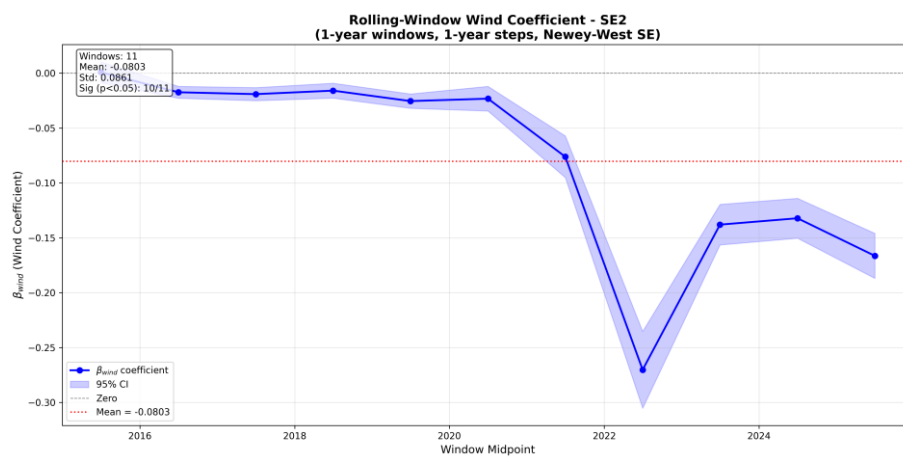


## SE2

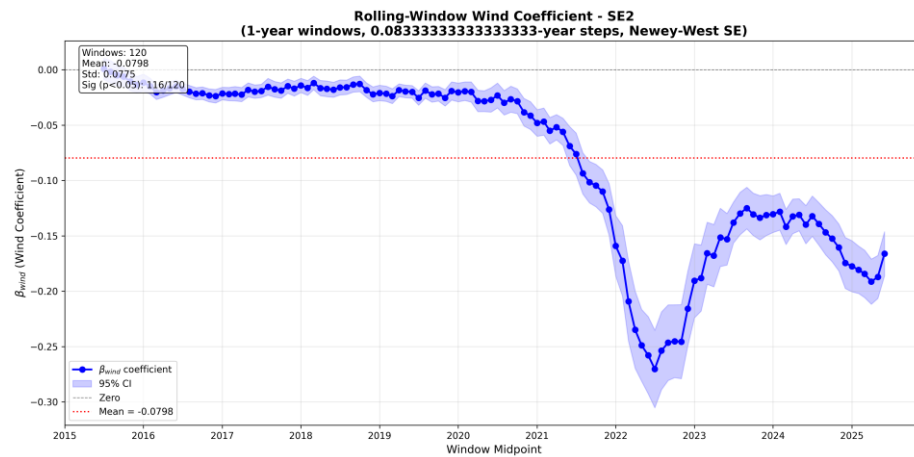
- 3 year, overlapping, shift 1 year



- 1 year, no overlapping, shift 1 year:

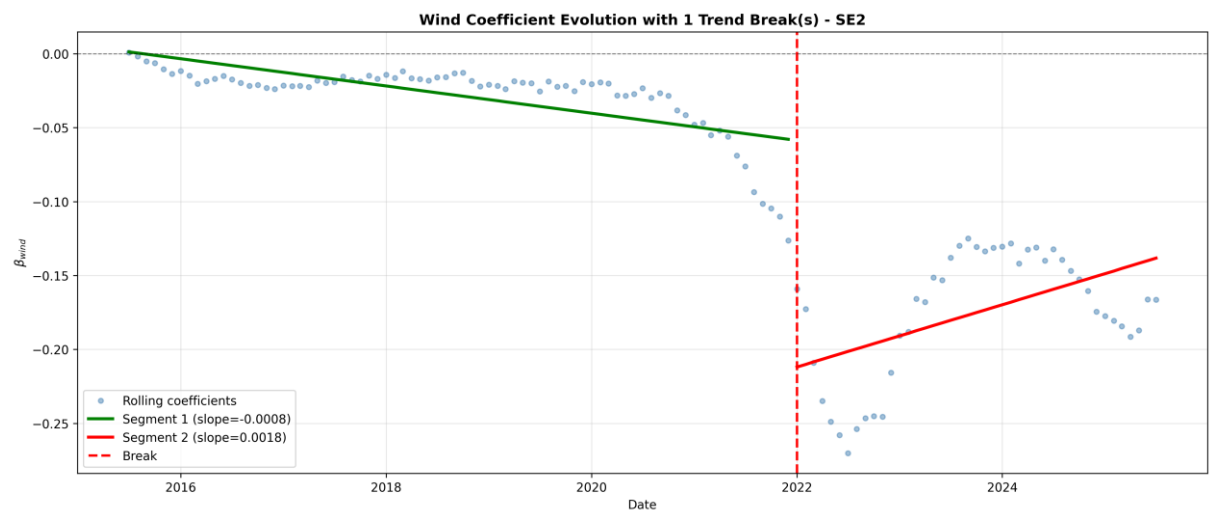


- 1 year, overlapping, shift by 1 month

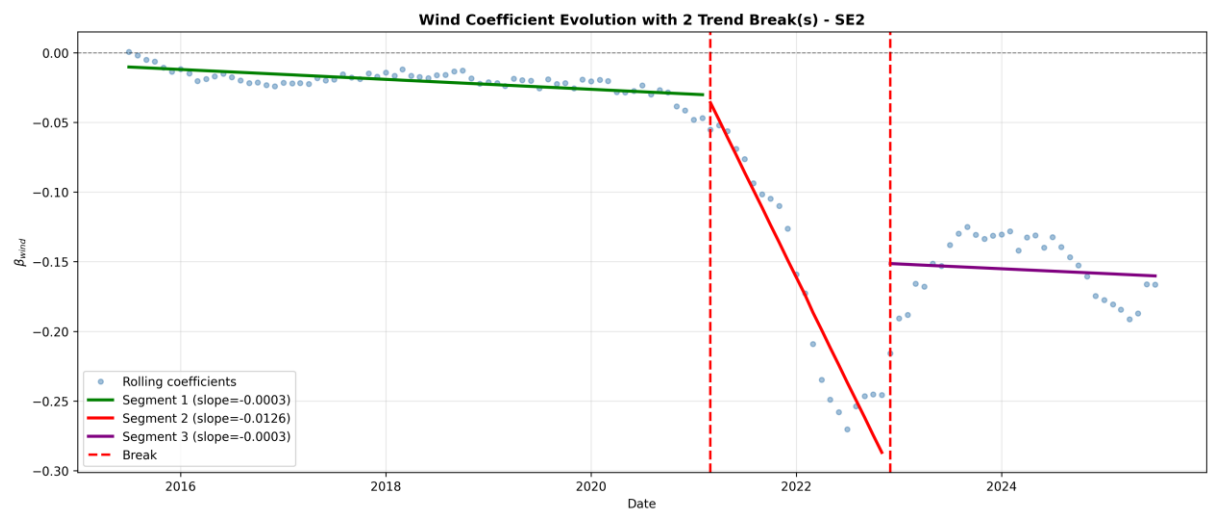


## Structural break analysis

- 1 break allowed

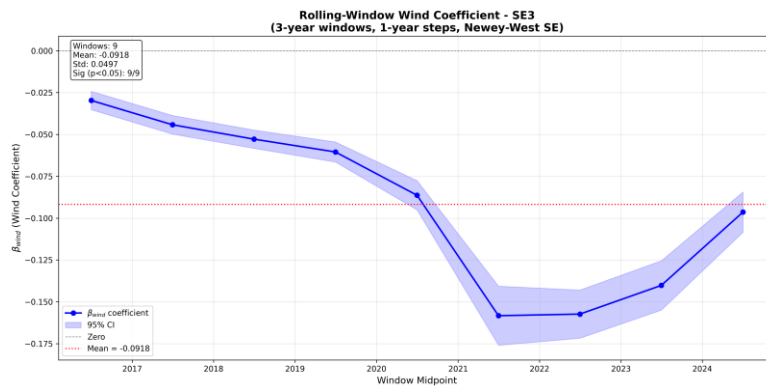


- 2 breaks allowed

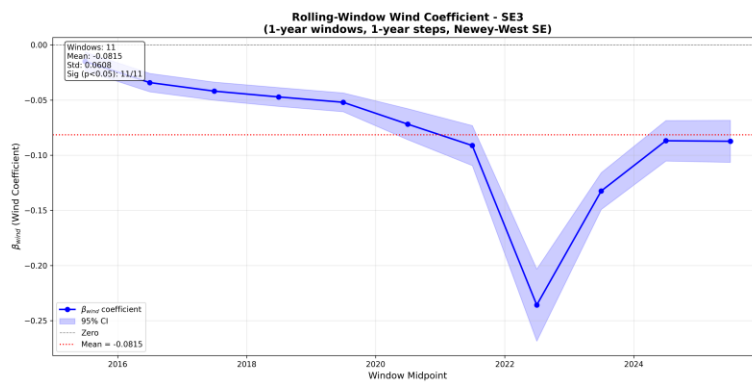


# SE3

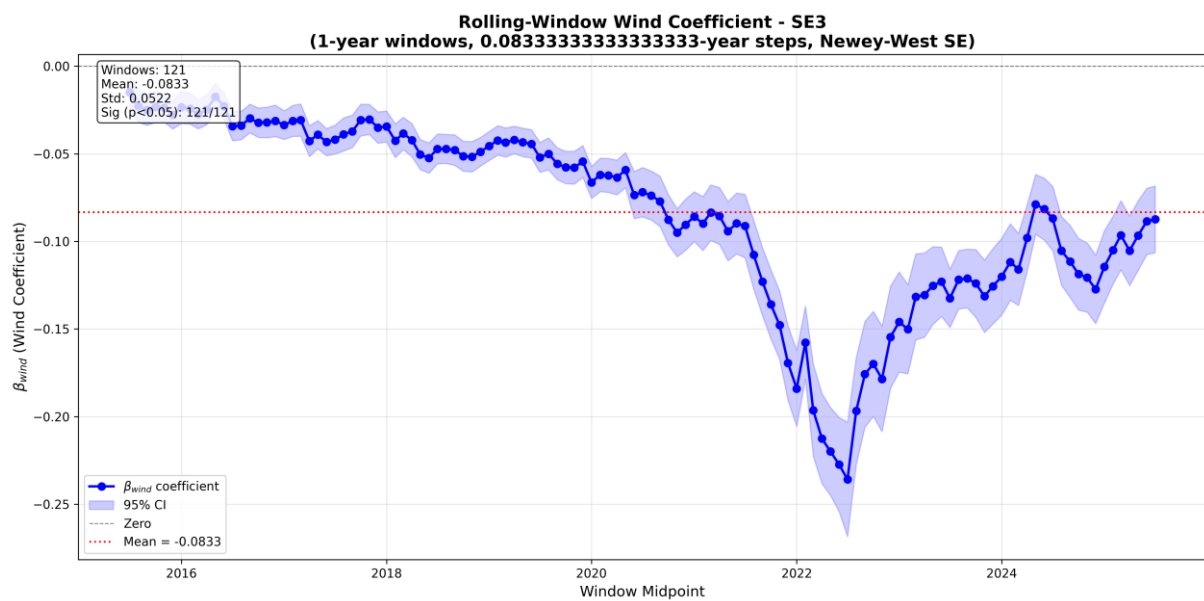
- 3 year, overlapping, shift 1 year



- 1 year, no overlapping, shift 1 year:

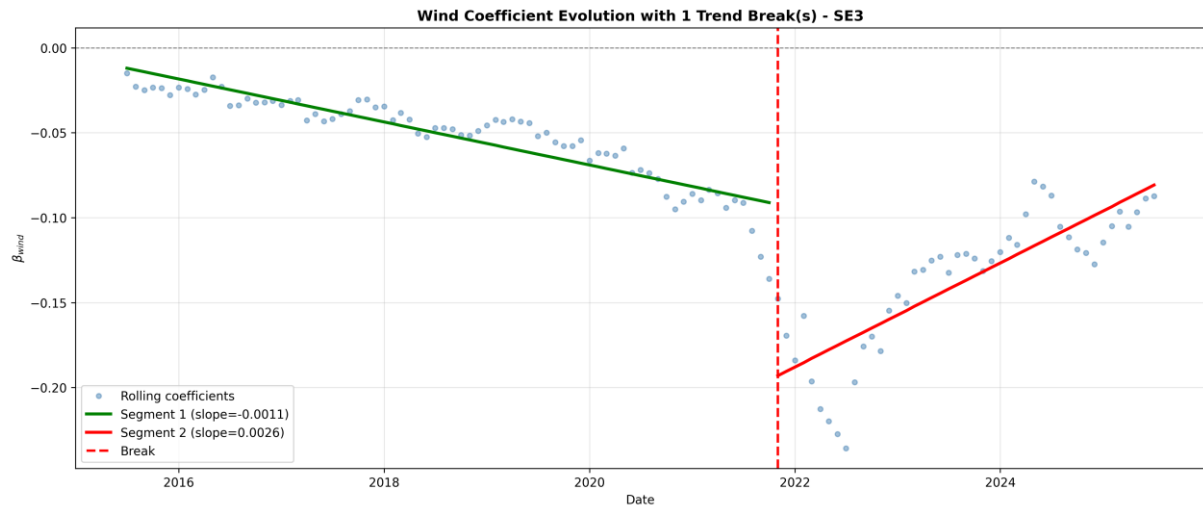


- 1 year, overlapping, shift by 1 month

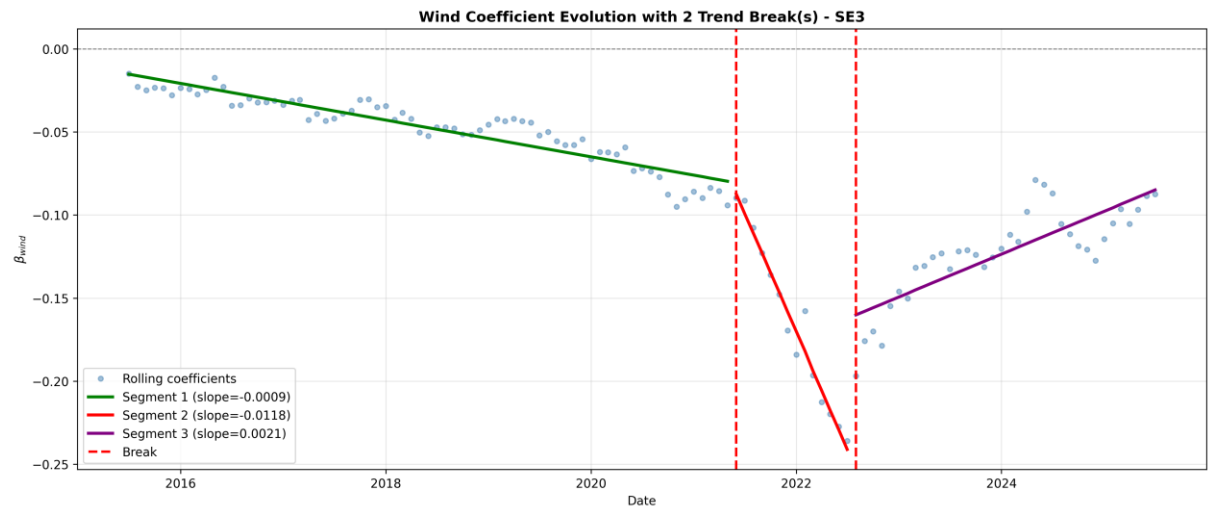


## Structural break analysis

- 1 break allowed

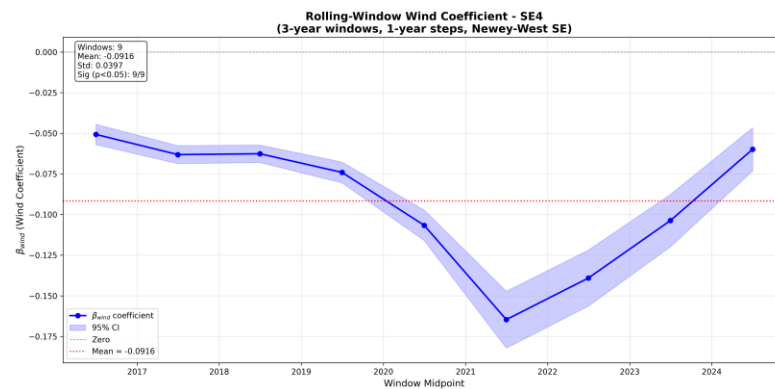


- 2 breaks allowed

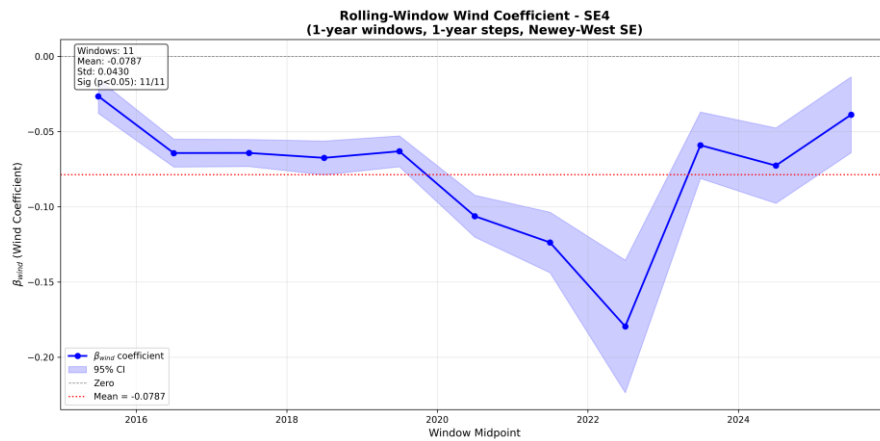


## SE4

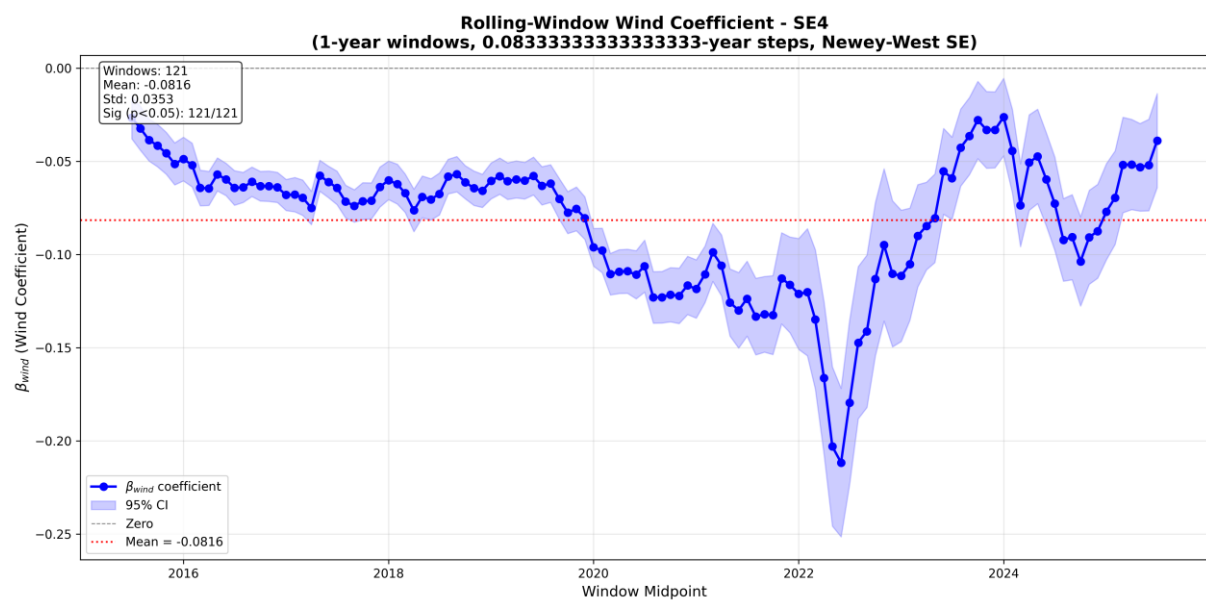
- 3 year, overlapping, shift 1 year



- 1 year, no overlapping, shift 1 year:

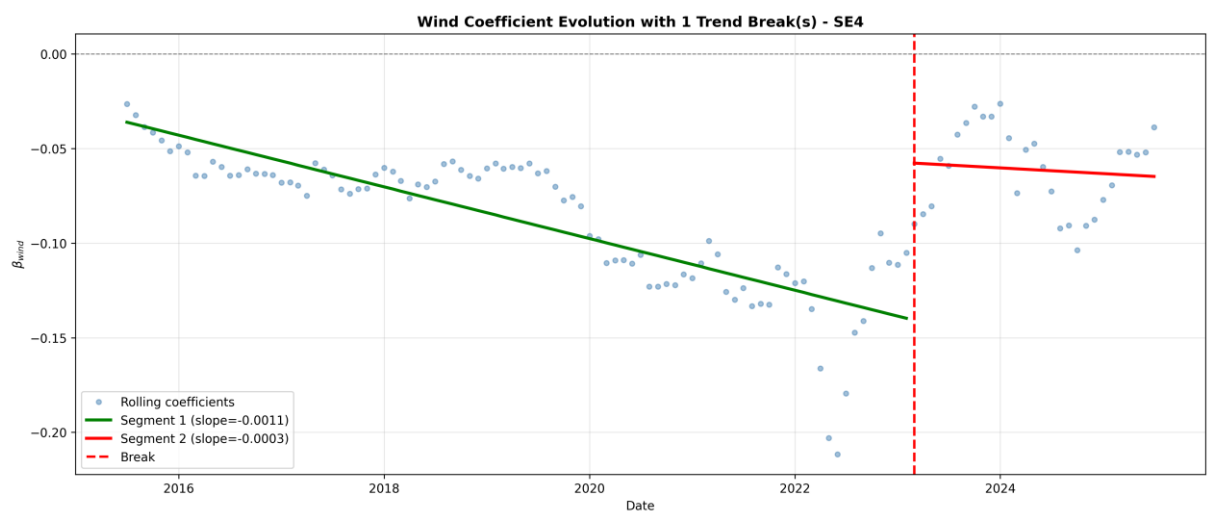


- 1 year, overlapping, shift by 1 month



## Structural break analysis

- 1 break allowed



- 2 breaks allowed

